Can Native MXFP4 Replace BF16 in FPGA Gated DeltaNet Decode? Cost and Long-Horizon Recurrent-State Drift

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***Abstract*—This paper asks whether native Open Compute Project (OCP) MXFP4 arithmetic can replace bfloat16 (BF16)-style arithmetic in a persistent-state Gated DeltaNet recurrence-core kernel on an AMD Alveo U55C field-programmable gate array (FPGA). Persistent state and the five-phase token schedule are inherited from prior work and held fixed. We implement lookup-table (LUT) E2M1 products, E8M0-aware integer accumulation, and a recurrence-aware two-term state with a three-entry write log (RS2/R3). Against single-precision floating point (FP32), the exact encoded path is evaluated at 64, 256, 1,024, 4,096, and 8,192 tokens. At token 8,192, BF16 reaches 0.999812 output cosine and 0.020998 recurrent-state relative L_2 error; uniform MXFP4 floating quantize/dequantize reaches 0.672818 and 0.903408, while native RS2/R3 reaches 0.996725 and 0.081526. The two registered long runs report zero element saturations, accumulator saturations, and scale clamps. RS2/R3 stores 55.02% of the BF16 state, but its HLS estimate uses 1.958 times the LUTs and 10.191 times the amortized cycles per STEP. Its all-layer out-of-context image fits, yet has -1.736 ns worst negative slack at 250 MHz; the first passing tested fixed-route point is 166.67 MHz. Thus native MXFP4 can satisfy the registered long-horizon synthetic stability bounds only with a mitigation that does not replace BF16 on the measured cost and timing axes. Four short model-derived Qwen layer-12 traces are diagnostic rather than closed-loop quality evidence, and power is vectorless rather than measured board energy.**

***Index Terms*—Gated DeltaNet, MXFP4, recurrent state, FPGA, numerical drift, Vitis HLS**

I. Introduction

Gated DeltaNet (GDN) replaces a growing attention history with a fixed-size recurrent matrix that is decayed and updated once per token [1]. This makes the numerical state persistent: an arithmetic error written at one token can affect every later token. Lowering state precision therefore changes more than storage cost. It changes the dynamical system implemented by the decoder.

Gupta et al. map GDN decode to a field-programmable gate array (FPGA) using persistent on-chip state and a five-phase BF16 pipeline [2]. We use that organization as the baseline dataflow. Neither persistent state nor the five-phase schedule is a contribution of this work. We study only the arithmetic replacement inside that recurrence-core boundary.

The Open Compute Project (OCP) MXFP4 format stores 32 E2M1 elements with one E8M0 scale [3]. Its 4.25 logical bits per value are attractive for a recurrent state that is read and rewritten every token. The shared exponent also creates two risks. Values below the block range disappear during alignment, and a scale change requantizes the complete block. Those errors feed back through the next state update.

We evaluate the replacement as a controlled experiment. The recurrence, logical K-by-V state, layer dimensions, U55C target, 4.0 ns constraint, $P_K = 16$, $P_V = 8$, and block size 32 are fixed. The study contributes:

1) an encoded-integer MXFP4 recurrence oracle and native high-level synthesis (HLS) arithmetic with explicit E2M1 decoding, E8M0 alignment, round-to-nearest-even (RNE), Q1.15 controls, and event counters;
2) long synthetic traces at 64, 256, 1,024, 4,096, 8,192 tokens for FP32, BF16, MXFP4 floating Q/DQ, native encoded RS2/R3, flat INT4, and an MXFP8-state fallback, plus four short model-derived recurrence traces and 6 scale refresh policies;
3) a fixed-geometry BF16, native-MXFP8, and RS2/R3 HLS comparison, with all-layer out-of-context post-route timing, utilization, design-rule, and vectorless-power evidence for all three implementations; and
4) an evidence-backed negative answer: recurrence-aware native MXFP4 meets the registered synthetic stability bounds and reduces state bytes, but loses to BF16 in LUT and cycle cost and misses the declared route-frequency target.

The evaluation covers one recurrence core, synthetic recurrent traces, four short model-derived layer-12 recurrence traces, matched HLS estimates, and out-of-context implementations. It does not measure Qwen3-Next perplexity, downstream accuracy, or board energy. Within that boundary, can native MXFP4 replace BF16-style arithmetic while reducing FPGA cost and limiting empirical recurrent-state drift over long synthetic decode traces?

II. Background and Scope

A. Microscaling Representation

An E2M1 element contains 1 sign bit, 2 exponent bits, and 1 mantissa bit. Its positive finite magnitude codebook is $\{0, 0.5, 1, 1.5, 2, 3, 4, 6\}$. Both signed-zero codes decode to zero, and the encoder emits positive zero. An E8M0 byte encodes the shared power-of-two scale 2^{e-127} for codes 0 through 254. Code 255 is invalid; an all-zero block uses scale code 127. NaN and infinity inputs are rejected. These rules follow the OCP representation [3]; scale selection and RNE at each state-write boundary are implementation choices.

B. Recurrence-Core Boundary

The head counts and K-by-V dimensions follow the pinned official Qwen3-Next-80B-A3B-Instruct configuration and Transformers implementation [4], [5]. For value head h, query/key head $p(h) = \lfloor h/2 \rfloor$ serves the corresponding pair of value heads. With logical state $S_h \in \mathbb{R}^{K\times V}$ and $K = 128$, $V = 128$, the token recurrence is

$$S_h^d = \alpha_h S_h, \tag{1}$$

$$u_h = \beta_h \left(v_h - k_{p(h)}^{\top} S_h^d \right), \tag{2}$$

$$S_h' = S_h^d + k_{p(h)} u_h^{\top}, \tag{3}$$

$$o_h = q_{p(h)}^{\top} S_h'. \tag{4}$$

The query is normalized and scaled by $1/\sqrt{128}$ before entering the kernel; the key is normalized. Gated RMSNorm, the z branch, projections, depthwise convolution, and output projection are outside the measured boundary. We therefore call the implementation a GDN recurrence-core kernel, not a complete GDN layer. Holding the unchanged convolution, projections, and output normalization outside the boundary isolates arithmetic-induced drift and cost in the persistent recurrence rather than mixing them with unmodified layer components.

The operation order is decay, prediction from the decayed state, delta formation, rank-one write, and output from the updated state. The software oracle was checked against pinned Transformers and FLA recurrent, chunk, and cache paths before quantized experiments were admitted [5], [6].

C. Prior Work and Component Ownership

DeltaNet establishes the rank-one erase/write recurrence and parallel training formulations on which GDN builds [7]; the gated recurrence itself is inherited from Yang et al. [1]. The persistent-state FPGA organization, paired-head handling, and five phases are inherited from Gupta et al. [2]. MXFP4-B32 is standardized [3]. QLlama maps microscaling quantization to FPGA inference, while Jack of All Scales studies MXFP dot-product mappings and digital signal-processing (DSP) block support on Agilex-5 [8], [9]. MXFormer implements block-exponent-aligned MXFP4 compute in charge-trap memory, MicroMix mixes MXFP4/6/8 channels, and MXAttention and an MXFP post-training-quantization benchmark identify scaling, clipping, and underflow as material quality factors [10], [11], [12], [13]. These are precedents for alignment, mixed precision, and scale-policy studies. Zhang et al. quantize the chunk-wise GDN matrix-inversion path rather than token-serial recurrent state [14]. KIVI retains recent full-precision entries beside a quantized cache, and SSDi8 uses a persistent INT8 state-space path with error correction [15], [16]. These works establish prior art for the component ideas. Our scope is the matched arithmetic-replacement experiment and its verification, not the invention of those constituents. Table I makes the resulting ownership boundaries explicit.

TABLE I
COMPONENT OWNERSHIP RELATIVE TO THE CORRECTED IMPLEMENTATION.

Component	Classification	Basis
Persistent on-chip state	Inherited	Gupta et al. [2]
Five-phase token schedule	Inherited	Gupta et al. [2]
Qwen recurrence and core boundary	Corrected	Pinned software parity
OCP MXFP4-B32 state	Modified	Standard format in a recurrent path
E2M1/E8M0 FPGA arithmetic	Modified	OCP format plus QLlama and Jack of All Scales [3], [8], [9]
Encoded arithmetic oracle	Newly proposed	Artifact-specific independent integer implementation
Two-term E2M1/E8M0 state and fixed R3 write log	Modified	Related residual-buffer methods exist [15], [16]; no first-use claim
Long-horizon drift harness	Newly proposed	Same recurrence and synthetic input prefix

Note: Newly proposed identifies ownership within this artifact, not a claim of literature priority.

III. NUMERICAL METHOD

A. Uniform MXFP4 Path

The matched BF16 comparator rounds state and operands to BF16 at their declared boundaries and performs reductions in FP32. The MXFP8-state fallback changes only the recurrent state to eight-bit E4M3 elements under E8M0 B32 scales while retaining MXFP4 token operands; flat INT4 uses signed symmetric four-bit quantization with a tensor-wide scale.

We report a floating Q/DQ MXFP4 diagnostic and a distinct native encoded path. The Q/DQ diagnostic quantizes at named boundaries but performs the intervening recurrence in floating point. It does not represent the HLS accumulator or its event behavior. The encoded path below is the software counterpart of the native HLS arithmetic.

The uniform path stores E2M1 elements in B32 blocks with an E8M0 scale per block. For nonzero block maximum m and largest finite E2M1 magnitude $x_{\max}$, the selected scale power is $s = \lceil \log_2(m/x_{\max}) \rceil$; its biased E8M0 code is clamped to the valid range. Elements are divided by 2^s and rounded to E2M1 with RNE. Zero has a canonical encoding. If E2M1 magnitude code r denotes $r/2$, a product has integer magnitude $r_a r_b$ and power $s_a + s_b - 2$. One flat 128-term reduction aligns every product directly to the greatest nonzero product exponent (power), applies RNE to the full exponent-difference shift, and accumulates in ascending key-index order with signed 32-bit saturation after every addition. A nonzero term rounded to zero is counted as an alignment underflow. This uniform baseline has no guard-bit offset; the five guard bits described below belong only to the corrected candidate.

Each synthetic STEP record supplies normalized q and k, value v, decay α, delta gate β, and the selected initial state S. The MXFP4 paths encode q, k, v, and every state write; α and β use unsigned Q1.15 fixed point. There are no projection weights in a test vector because all projections are outside the recurrence-core boundary. The MXFP8 fallback changes only state element codes to E4M3 under B32 E8M0 scales;

flat INT4 changes the same token operands and state under its declared tensor-wide scale. Requantization, element and accumulator saturation, scale clamps, alignment underflows, and scale changes are explicit state-write events. The complete contract is versioned in the artifact rather than inferred from C++ types.

The element product uses sign exclusive-or (XOR), table-based E2M1 magnitude multiplication, and exponent addition. No floating-point type appears in the active native arithmetic sources. The resulting E2M1 element-product logic is integer based; wider coefficient operations may still map to DSPs and are reported as such.

B. Recurrence-Aware Correction

Native encoded MXFP4 fails the frozen finite-trace thresholds, so we also evaluate one locally frozen correction. The base state is the sum of two E2M1 terms, each with an independent E8M0 B32 scale. Token q, k, and v vectors and each live rank-one write use the same two-term representation. Up to three writes remain in a log with Q1.15 decay coefficients. At capacity, all three entries are folded tilewise into the base and the log is cleared. The encoded implementation aligns shared exponents with 5 guard bits before signed 32-bit accumulation; these constants are emitted as generated macros and tables with the result artifacts.

This representation is a mitigation, not the controlled uniform replacement. It changes the physical state representation and adds a periodic fold. Its logical payload is 576,912 bytes per layer, 55.02% of BF16 and 6.70% larger than a uniform MXFP8-E4M3-B32 state. Exact arithmetic equivalence was tested separately before quantized evaluation.

IV. FPGA Implementation

Both controlled baselines use equivalent persistent-state command semantics and the same five-phase token organization. Commands select one of 36 runtime layer slots and support RESET, LOAD, STEP, and READBACK. Their data-plane encodings are not identical: BF16 transports BF16 tensors, whereas MXFP4 transports encoded elements and scales and returns expanded integer output fields. Packing and token quantization are outside both HLS kernels. Table II reports exact logical tensor payloads for LOAD, READBACK, and STEP. These are type-layout counts before interface packing, not measured AXI traffic; RESET has no tensor payload. The HLS target is `xcu55c-fsvh2892-2L-e` under Vitis HLS 2025.2 at 4.0 ns with $P_K = 16$, $P_V = 8$, and B32. We retain the U55C target used by the inherited BF16 dataflow so the arithmetic comparison does not also change devices. We fix this target and parallelism point for the replacement rather than tune each format independently.

The corrected candidate adds packed primary and residual banks, three packed key/update log entries, per-head gamma and lambda coefficients, and event counters. A non-mutating snapshot exposes all persistent fields for exact verification. Folding is atomic at the command boundary.

TABLE II
Logical command payload sizes before AXI packing.

Logical payload	BF16	Native MXFP4 RS2/R3
LOAD/READBACK state (bytes/layer)	1,048,576	576,912
STEP input (bytes)	16,512	8,832
STEP output (bytes)	8,192	24,576

Figure 1 maps the correction onto the inherited five-phase schedule. The base state remains resident while each STEP decays its coefficient, evaluates the base and live writes, appends the new rank-one update, and reads the output from the updated logical state. When the three-entry log fills, a tilewise fold runs after the output has observed the newest write.

Logically, B32 blocks run along the V-column axis within each K row. The code uses the same P_K and P_V constants as the BF16 comparator and targets an initiation interval (II) of 1 on its explicitly pipelined inner loops. All reported targeted loops meet II=1. HLS estimates 321.96 MHz and reports 167,082 LUTs and 26 DSPs. The DSPs implement coefficient and wider arithmetic; the E2M1 element multiplier itself remains integer logic. Section VI-D separately reports the physically allocated 36-slot out-of-context implementation.

V. Evaluation

A. Questions and Controls

The experiments answer these questions:

- Does native encoded MXFP4 limit empirical output and recurrent-state drift over 8,192 synthetic tokens relative to FP32 and BF16?
- Do floating-Q/DQ scale refresh policies or an MXFP8-E4M3 state fallback meet the same finite-trace criteria?
- If a recurrence-aware correction meets the finite-trace thresholds, does its HLS estimate improve on the fixed-geometry BF16 kernel?
- Does that corrected candidate physically fit and close the declared clock target after out-of-context routing?

The controlled variants use 16 query/key heads, 32 value heads, $K = 128$, $V = 128$, the same K-by-V state, and the same generated input prefix. The development seed is `0xFB72`. Required checkpoints are 64, 256, 1,024, 4,096, 8,192. The trace draws query and key vectors from independent standard normal distributions before normalization. Values and random initial states are zero-mean normal with standard deviations 0.25 and 0.05, respectively. Nominal decay and delta gates follow $\mathcal{U}[0.950, 1.000]$ and $\mathcal{U}[0.000, 1.000]$. The long-horizon comparison uses the high-retention family, which keeps the vector distributions but uses $\alpha \sim \mathcal{U}[0.995, 1.000]$ and $\beta \sim \mathcal{U}[0.850, 1.000]$, with both random and zero initial states. Changing the trace-family tag changes the deterministic RNG stream, even at the same seed. Every variant in the main checkpoint table consumes the same high-retention random-state input prefix; zero-state and held-out runs are additional RS2/R3 stability checks. The frozen engineering gate requires output cosine at least 0.99 at every checkpoint,

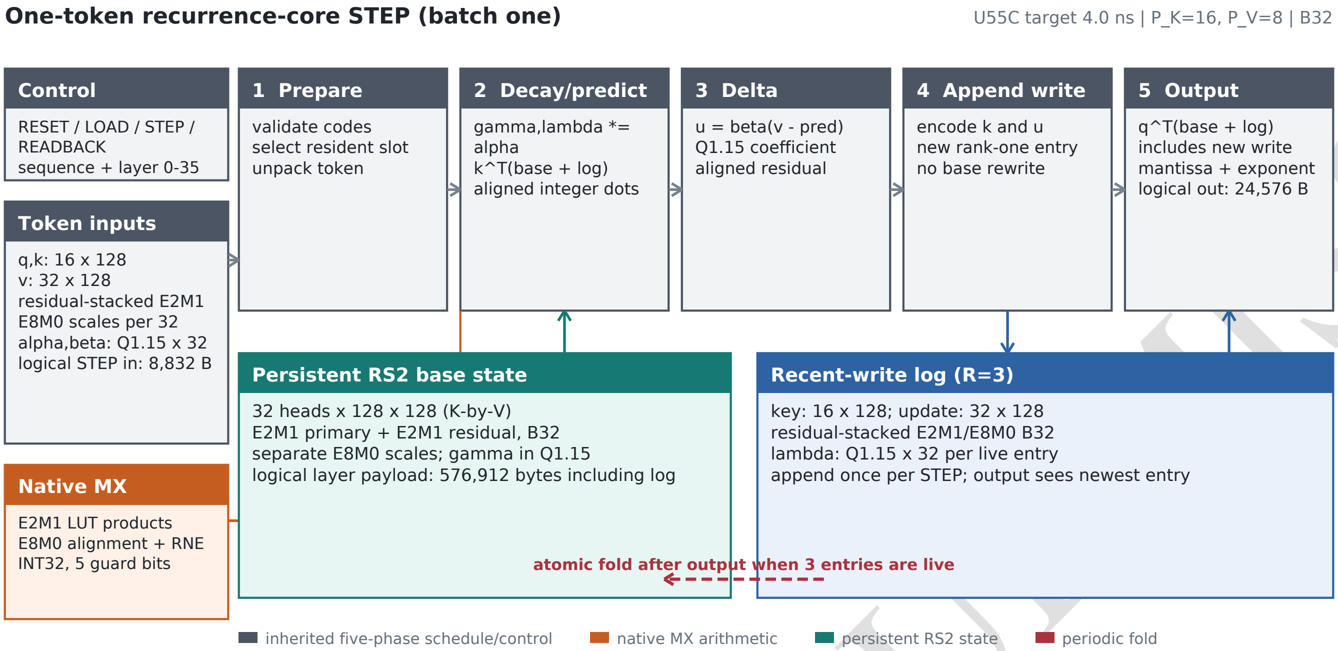

Fig. 1. Corrected-candidate datapath on the inherited five-phase token schedule. Gray denotes inherited control and scheduling, orange denotes native MX arithmetic, teal and blue denote the corrected persistent representation, and the dashed path is the periodic atomic fold. Dimensions and rates are generated from the numerical contract; the drawing is logical architecture, not physical-placement evidence.

final state relative L_2 at most 0.10, and zero element saturation, accumulator saturation, and scale-clamp events. These are the three hard-event counters; alignment underflows and state-scale changes are reported but excluded from the zero-event condition. We fixed these complementary diagnostics before test-set access: cosine constrains observable output direction, while state relative L_2 constrains hidden-state magnitude. They were not calibrated to downstream model quality.

RS2/R3 uses 6 held-out conditions formed by crossing the registered seed blocks with random and zero initial states. Every artifact was deterministically recomputed by the same implementation and compared field by field with its recorded comma-separated values (CSV) artifact. This checks integrity and determinism, not implementation independence. The 2 8,192-token development runs use the same recorded seed with random and zero initial states. Each longer run was fully recomputed from source and compared field by field with its recorded CSV artifact. The encoded HLS candidate was also checked against frozen arithmetic cases and a stateful C-simulation trace through repeated folds and final snapshot readback.

The corrected candidate is synthesized and routed out of context for the declared U55C part. A source-level clock-buffer insertion is hash-checked against the HLS-generated top before implementation. Timing is re-reported on the same routed checkpoint at several clock periods; the first passing tested point is not interpreted as a binary-searched maximum frequency. Post-route power uses Vivado vectorless activity propagation at a timing-closed 6.0 ns period. It is a tool estimate, not board telemetry. Vitis HLS/XSIM also runs 2 exact early-return commands through the generated RTL. Those commands exercise control/status behavior but no recurrent state transition. A direct AXI harness over the same HLS-generated Verilog executes 66 ordered commands and compares 262,144 output fields plus the final persistent-state snapshot. Its generated status macro is PASS. Official full-trace XSIM is kept as a separate evidence row rather than inferred from the direct harness.

Separately, a pinned real Qwen3-Next layer-input capture at layer 12 is combined with the matching checkpoint projection weights to reconstruct query, key, value, decay, and delta-gate inputs at the post-convolution, pre-query/key-normalization recurrence boundary. The 4 prompt traces contain 60 valid tokens in total, with 12–18 tokens per prompt. They are evaluated as floating Q/DQ diagnostics and are not mixed into the synthetic long-horizon curves. This is a short real-input recurrence check, not closed-loop quantized model inference.

B. Metrics and Evidence Levels

Output cosine similarity flattens all value heads and V coordinates at each token. State relative L_2 flattens all heads, K rows, and V columns and is $\|S_q - S_{32}\|_F / \max(\|S_{32}\|_F, 1 \times 10^{-12})$; maximum absolute state error and arithmetic events are also recorded. Exact equality returns cosine one and zero error; the artifact specifies the near-zero-norm cosine convention. No headwise minimum, percentile, or acceptance gate is computed. Software traces, HLS C simulation, HLS synthesis, register-transfer-level (RTL) simulation, post-route implementation, and board measurement are separate evidence levels. A PASS at one level does not fill a missing later level.

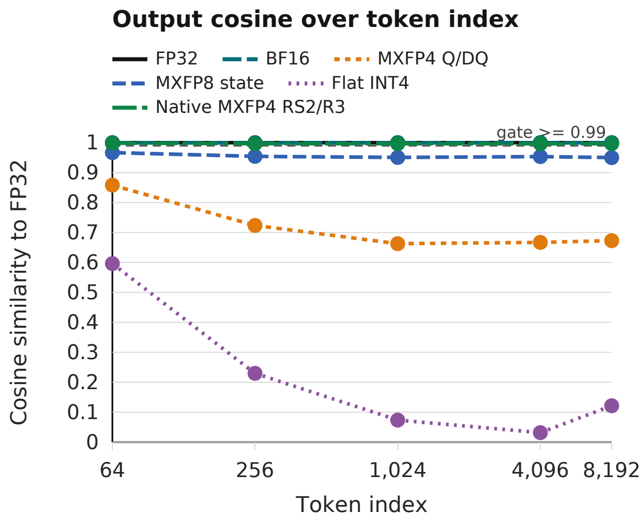

Fig. 2. Output cosine versus token index on the controlled synthetic trace. The threshold is an engineering criterion, not a model-quality guarantee.

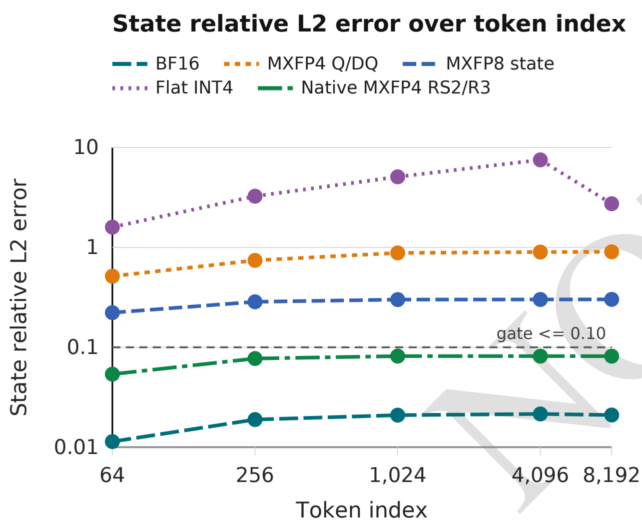

Fig. 3. Recurrent-state relative L_2 error versus token index on the same trace. All variants share the recurrence, dimensions, and input prefix.

VI. Results

A. Uniform Arithmetic Replacement

Figures 2 and 3, together with Table III, use one controlled high-retention input prefix for every variant. BF16 remains close to FP32 through the final checkpoint. Uniform MXFP4 floating Q/DQ and flat INT4 drift substantially. Keeping MXFP4 token operands while widening only the state to MXFP8 improves both metrics but does not meet the registered RS2/R3 state-error bound. Native RS2/R3 reaches 0.996725 output cosine and 0.081526 state relative L_2 at token 8,192, while BF16 reaches 0.999812 and 0.020998. The uniform Q/DQ curve is a diagnostic and is not paired with the encoded RS2/R3 hardware cost as if they were one method.

a) Real-capture checks.: The pinned Qwen layer-input capture contains 147,456 values. Its absolute-value 99th and 99.9th percentiles are 0.6836 and 2.5781, while the maximum is 51.25, or 75.0 times the 99th percentile. This confirms that a real layer boundary can contain a much wider tail than the nominal Gaussian trace. The reconstructed recurrence traces in Table IV give BF16 mean final output cosine/state relative L_2 of 0.999991/ 0.005894, MXFP4 gives 0.982048/ 0.324788, MXFP8 state gives 0.989173/ 0.207192, and INT4 gives 0.866071/ 0.774523. The exact RS2/R3 path gives 0.999635/ 0.047009 on average; its worst cosine and state relative L_2 are 0.999408 and 0.050485. It records 2 transient accumulator saturations. These short traces cannot establish long-horizon stability, perplexity, or downstream accuracy.

No tested floating-Q/DQ scale policy rescues uniform MXFP4. Fixed scales retain more output direction but produce extensive state-element clipping; frequent refresh reduces clipping without satisfying the drift gate. Periodic and threshold-triggered refresh also fail. Table V reports the complete token-8,192 ablation rather than selecting one favorable metric.

B. Fixed-Geometry HLS Cost

Table VI holds the recurrence, dataflow, target, parallelism, and block size fixed. RS2/R3 reduces logical state bytes but uses 1.958 times the BF16 LUT estimate, 3.200 times the maximum non-fold STEP cycles, and 10.191 times the amortized cycles once periodic folds are charged. Native MXFP8 is synthesized at the same boundary as the wider low-precision comparator. The different data-plane encodings and excluded packing mean these ratios are not end-to-end interface or energy ratios.

Figure 4 separates three non-interchangeable axes: logical state payload, final synthetic quality, and HLS cycle estimate. The state totals cover one sequence of 36 recurrence cores; they exclude weights, convolution state, full-attention key/value caches, activations, shell logic, and scratch space. They therefore answer whether the declared recurrence bank fits, not which complete model fits on the card. Matched INT4 hardware remains unrun and is not imputed.

C. Registered RS2/R3 Stability

RS2/R3 passes the frozen held-out protocol for all 6 runs and both initial-state conditions. Table VII reports the worst held-out metric beside the two independently recomputed 8,192-token development runs. The long runs reach a minimum all-token cosine of 0.995974 and maximum all-token state relative L_2 of 0.082737, with 0 element saturations, 0 accumulator saturations, and 0 scale clamps. Alignment underflow remains a counted diagnostic because discarding terms below the retained integer window is part of the declared arithmetic; it is not silently reclassified as saturation. These are finite-trace engineering checks, not a formal stability certificate.

D. Post-Route and Generated-RTL Evidence

All three matched recurrence cores physically fit in the U55C out-of-context flow. Table VIII reports their routed

TABLE III

Controlled synthetic drift at every required checkpoint. Each cell gives output cosine/state relative L_2 against FP32.

Variant	64	256	1,024	4,096	8,192
FP32 reference	1.000/0.000	1.000/0.000	1.000/0.000	1.000/0.000	1.000/0.000
BF16 operands/state	1.000/0.011	1.000/0.019	1.000/0.021	1.000/0.022	1.000/0.021
MXFP4 floating Q/DQ	0.858/0.515	0.724/0.742	0.663/0.880	0.667/0.898	0.673/0.903
MXFP4 + MXFP8 state	0.967/0.222	0.954/0.285	0.951/0.300	0.954/0.301	0.950/0.302
Flat INT4	0.596/1.595	0.230/3.253	0.074/5.088	0.032/7.519	0.121/2.737
Native MXFP4 RS2/R3	0.998/0.054	0.997/0.077	0.997/0.082	0.997/0.082	0.997/0.082

TABLE IV

Exact RS2/R3 and floating-Q/DQ recurrence diagnostics on four model-derived Qwen layer-12 prompt traces. Final values are averaged per prompt; "worst" and maximum absolute error range over all prompts and valid token positions.

Arithmetic	Mean final cos.	Worst cos.	Mean state L_2	Worst state L_2	Max abs.	Nonfinite
BF16 state/operands	0.99999	0.99998	0.00589	0.00657	0.01917	0
MXFP4 state/operands	0.98205	0.96768	0.32479	0.33888	0.61595	0
MXFP4 operands + MXFP8 state	0.98917	0.96967	0.20719	0.21994	0.46683	0
Flat INT4	0.86607	0.75192	0.77452	0.82253	0.40506	0
Native MXFP4 RS2/R3	0.99963	0.99941	0.04701	0.05049	0.07849	0

TABLE V

Uniform-MXFP4 block-scale policies at token 8,192. Clips and scale changes are cumulative floating quantize/dequantize diagnostics.

Scale policy	Output cosine	State rel. L_2	Clips	Scale changes
Fixed	0.847089	0.676620	67,980,414	0
Every token	0.729275	1.103841	73	26,504,191
Periodic, $N = 4$	0.773418	0.985345	16,091,670	8,345,806
Periodic, $N = 8$	0.785090	0.951738	20,026,426	4,448,732
Periodic, $N = 16$	0.785347	0.939046	22,646,994	2,354,947
Threshold triggered	0.558410	1.724816	0	35,057

TABLE VI

Fixed-geometry U55C HLS logic and schedule estimates.

Metric	BF16	Native MXFP4 RS2/R3	Ratio
Logical state bytes/layer	1,048,576	576,912	0.550
HLS LUT	85,350	167,082	1.958
Non-fold STEP max cycles	4,225,856	13,522,144	3.200
Amortized cycles/STEP	4,225,856	43,065,280	10.191
Estimated Fmax (MHz)	342.47	321.96	–
Explicit II=1 loops	PASS	PASS	–

TABLE VII

Registered RS2/R3 finite-trace stability. Rows aggregate independent random- and zero-state runs; all metrics are measured against FP32.

Evidence set	Runs	Min. cosine	Max. state rel. L_2	Max. final rel. L_2	Max. abs. error
Held-out, 1,024 tokens	6	0.996099	0.082666	0.081983	0.142368
Development, 8,192 tokens	2	0.995974	0.082737	0.081702	0.162014

resources, first passing tested frequencies, and vectorless power. RS2/R3 uses 82,038 CLB LUTs, 341 BRAM tiles, 598 URAMs, and 12 DSPs. Its design-rule check has no critical warning or error, but physical fit does not imply target timing: WNS is -1.736 ns at 250 MHz, and the first passing point in the fixed-route sweep is 166.67 MHz. A post-route aggressive-fanout pass improves slack slightly; subsequent retiming, SLR-crossing, general Explore, AggressiveExplore, and a chained fanout-plus-AggressiveExplore pass also fail the declared target. The best WNS remains strongly negative, so further timing work requires architecture-level or floorplanning changes rather than another post-route directive. We retain the original routed checkpoint for the matched table. An isolated experiment then registered the four URAM output paths. It preserved exact C-simulation and all explicit synthesis loop constraints, but left the HLS clock estimate unchanged while increasing logic and worst-case latency. Because the routed failures terminate on URAM enable and write-control pins, the output-latency variant does not target the failing path and was not promoted or routed. A second isolated variant moved the fold decision into a non-inlined helper. It preserved exact C-simulation and the HLS schedule, and its matched route reduced control fanout and setup violation. The target still failed, however; the remaining worst path runs from the local fold state machine to a URAM byte-write-enable pin. We therefore retain this as architecture-diagnostic evidence rather than replacing the source-locked candidate. A third isolated

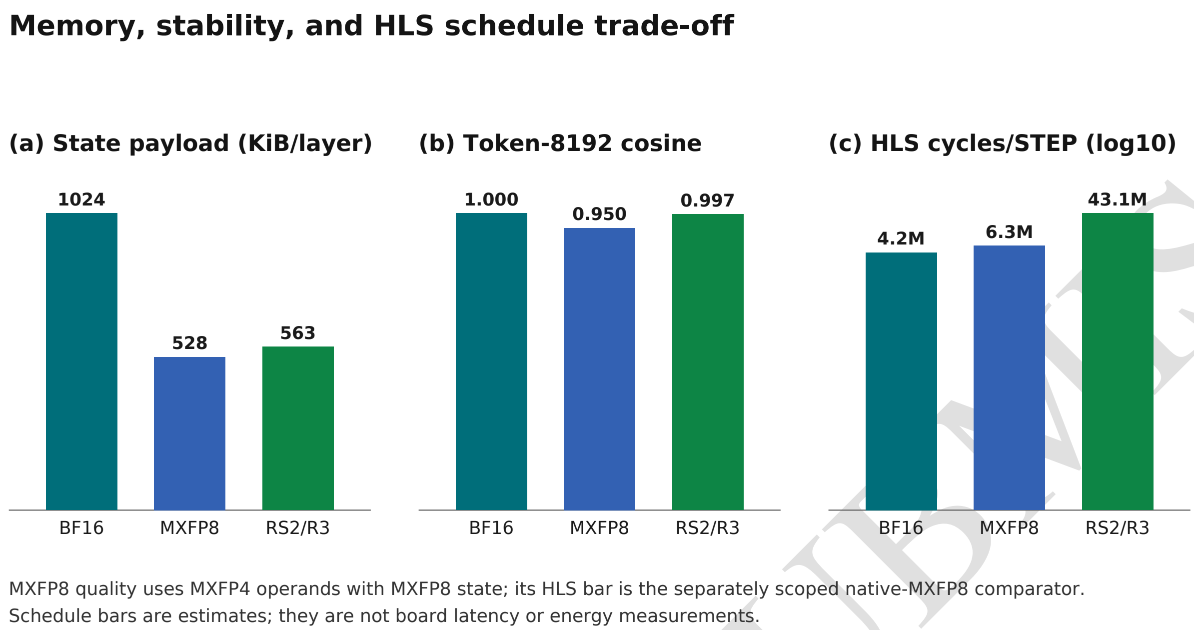

Fig. 4. Memory–performance–accuracy view for the controlled comparators. It combines logical storage, synthetic quality, and HLS estimates, not board energy or a measured end-to-end Pareto frontier.

variant retained that fold boundary and moved each folded-state write into a separate non-inlined commit stage. It again preserved exact C-simulation and the HLS loop constraints, and produced the strongest isolated setup improvement with fewer failing endpoints. It nevertheless missed the declared clock, increased periodic-fold latency, and moved the worst high-fanout path back to outer fold control driving a resident-primary URAM enable. This variant is likewise not promoted. A fourth isolated variant completely partitioned the primary and residual state stores by layer. This preserved the exact recurrent trace and all-layer capacity, and it removed the monolithic high-fanout write-enable path. The routed design instead exposed a bank-local read/selector path into fold logic, increased LUT and DSP cost, and worsened setup slack, total negative slack, and the failing endpoint count. We therefore reject simple complete layer banking as a timing fix; bank locality must be co-designed with layer selection and read-return pipelining. A fifth isolated variant instead used factor-six cyclic layer banking to reduce the selector and logic cost of complete partitioning. It preserved the exact recurrent trace, all explicit synthesis loop constraints, and all-layer capacity. The matched route slightly improved worst setup slack relative to the selected design, but still missed the target while worsening total negative slack and the failing-endpoint count and increasing LUT, register, and DSP use. Its worst path again crosses an SLR into high-fanout URAM control. We therefore reject partial layer banking as a standalone timing fix and retain the source-locked candidate. Finally, we composed factor-six banking with the strongest isolated fold-write hierarchy. Exact C-simulation and every explicit synthesis constraint still passed, but the matched route was worse than either parent in worst slack, total negative slack, and failing endpoints. The critical path crossed two SLRs and remained dominated by interconnect into resident-state memory. This negative composition result shows that reducing one high-fanout fold-control net does not solve the wider snapshot, reset, and bank-selection distribution problem. We reject the composition and leave the source-locked candidate unchanged. A seventh isolated variant placed the remaining snapshot-load state writes behind a non-inlined commit boundary on top of that composition. Exact C-simulation, every explicit synthesis constraint, routing, hold, and design rules passed. The boundary removed snapshot-load control from the worst-path startpoints and modestly improved worst slack, but total negative slack and the failing-endpoint count grew substantially. The worst paths instead originated primarily in shared layer/address control, with smaller contributions from resident-memory clocks, top-level control, and fold control. We therefore retain snapshot-write locality only as a clue for a lower-cost unbanked experiment; this composed variant is not promoted. An eighth isolated variant returned to the strongest unbanked fold-write parent and split its combined primary and residual commit into sequential non-inlined helpers. Exact C-simulation, every explicit synthesis constraint, routing, hold, and design rules passed, while HLS resource use changed negligibly. The matched route nevertheless regressed sharply in worst slack, total negative slack, and failing endpoints. Snapshot-load, fold, reset, and top-level control accounted for every worst-path startpoint, and the longest path crossed two SLRs with delay dominated by interconnect. Separating one

TABLE VIII
MATCHED OUT-OF-CONTEXT PHYSICAL RESULTS. POWER IS VECTORLESS; EACH FREQUENCY IS THE FIRST PASSING POINT IN THE TESTED FIXED-ROUTE SWEEP.

Metric	BF16	Native MXFP8	MXFP4 RS2/R3
Physical fit	PASS	PASS	PASS
First tested closing MHz	140.35	166.67	166.67
CLB LUT	57,942	58,582	82,038
FF	38,550	39,807	69,800
BRAM tiles	1602.5	164.5	341.0
URAM	928	576	598
DSP	15	2	12
Vectorless total power (W)	5.648	4.952	5.242

TABLE IX
VIVADO VECTORLESS POWER COMPONENTS AT EACH VARIANT'S FIRST PASSING TESTED CLOCK POINT. THESE ARE TOOL ESTIMATES, NOT BOARD MEASUREMENTS.

Component (W)	BF16	Native MXFP8	MXFP4 RS2/R3
CLB logic	0.230	0.218	0.335
Signals	0.487	0.380	0.450
Clocks	0.306	0.272	0.294
BRAM	0.364	0.117	0.137
URAM	0.836	0.605	0.654
DSP	0.009	0.002	0.004
Dynamic total	2.231	1.593	1.874
Static	3.416	3.359	3.368
On-chip total	5.648	4.952	5.242

write-control cone therefore displaced rather than removed the global resident-memory control problem; the split variant is rejected and not promoted. The ninth variant applied snapshot-write locality directly to that unbanked parent. Exact C-simulation and physical checks passed, and snapshot write/load logic left the worst-path origins, but setup regressed as top-level, URAM-read, and fold control became dominant. Snapshot-write isolation alone is rejected. The tenth variant instead added two contiguous layer banks to the fold-write parent. Exact C-simulation and physical checks again passed, but setup regressed sharply: 80 of the 100 worst paths began in the state-input AXI load FIFO and crossed two SLRs into banked URAM control. We reject contiguous banking without a separately pipelined load/control plane. An eleventh variant combined those banks with snapshot-write locality. It removed load and snapshot startpoints, but all 100 worst paths moved to the top FSM and timing worsened further. We reject this composition; closing 250 MHz requires a redesigned memory-owner/control hierarchy rather than another local helper boundary.

The direct Verilator harness drives the HLS-generated Verilog through snapshot LOAD, 66 commands, all log-fold boundaries, and final READBACK. It compares 262,144 decoded output values plus counters and the complete final encoded state against the independent integer oracle; the 64-token PASS result establishes recurrent generated-RTL parity for the required trace. Official XSIM control smoke is reported separately and is not substituted for the recurrent run.

At the timing-closed 6.0 ns point, RS2/R3 has a 5.242 W vectorless estimate, split into 1.874 W dynamic and 3.368 W static power. Table IX gives the same report categories for every routed variant. Because activity is vectorless and no board telemetry is available, we do not derive energy per token or an energy-efficiency ratio.

VII. DISCUSSION

The controlled answer is no for the current implementation. Uniform MXFP4 drifts substantially, and widening only the recurrent state to MXFP8 improves but does not remove that drift. RS2/R3 restores the registered finite-trace criteria and reduces logical state bytes relative to BF16, but it is not a cost-reducing arithmetic replacement: its HLS LUT estimate, non-fold latency, amortized latency, and routed target-frequency result all lose to the BF16 baseline.

This result separates representation density from implementation cost. Shared scales reduce bits, but exponent comparison, alignment, requantization, and folding add control and arithmetic. The E2M1 multiplier is small and DSP-free; the surrounding recurrence is not. The fixed write log also creates a periodic fold whose service cost dominates the amortized schedule. A useful replacement must simplify the complete state-update path rather than only narrowing the stored elements or replacing a multiplier.

The word *drift* is deliberate. For one head, let E_t be candidate minus reference state. For the same external token inputs, define R_t as the exact residual between the implemented candidate transition and the ideal reference transition applied to the candidate state. This residual includes input and state quantization, alignment, accumulation, scale, clipping, and fold error. The recurrence then gives

$$E_t = \alpha_t (I - \beta_t k_t k_t^{\top}) E_{t-1} + R_t. \tag{5}$$

For normalized k_t and $\beta_t \in [0, 1]$,

$$\|E_t\|_F \leq \alpha_t \|E_{t-1}\|_F + \|R_t\|_F. \tag{6}$$

Let $P_t = k_t k_t^{\top}$. The parallel and orthogonal components obey the exact identity

$$\begin{aligned} \|(I - \beta_t P_t)E\|_F^2 = (1 - \beta_t)^2 \|k_t^{\top} E\|_2^2 \\ + \|(I - P_t)E\|_F^2. \end{aligned} \tag{7}$$

Thus the delta gate attenuates only error parallel to the current key; the worst-case operator norm is one because $K > 1$. Unrolling the one-step bound gives

$$\|E_T\|_F \leq \Big(\prod_{i=1}^{T} \alpha_i\Big) \|E_0\|_F + \sum_{j=1}^{T} \Big(\prod_{i=j+1}^{T} \alpha_i\Big) \|R_j\|_F. \tag{8}$$

If $\alpha_i \leq \bar{\alpha} < 1$ and $\|R_i\|_F \leq \bar{R}$, the right-hand side is at most $\bar{\alpha}^T \|E_0\|_F + (1 - \bar{\alpha}^T)\bar{R}/(1 - \bar{\alpha})$. Our protocol permits decay arbitrarily close to one, so it supplies no uniform contraction margin. Changing keys may attenuate different directions, but a persistent-excitation bound is not established. These relations explain why repeated local errors matter, but they are not a formal stability certificate. We do not have a conservative bound on R_t spanning input and base quantization, logged coefficients, folding, alignment, accumulation, scale changes, and clipping. "Stable" in the engineering gate therefore means only that the measured finite traces stay within its thresholds.

The correction also changes the experimental role of MXFP8. MXFP8 state is a stronger numerical baseline than uniform MXFP4, and the RS2/R3 payload is 6.70 percent larger than uniform MXFP8. All three matched images fit out of context, but none of the vectorless power estimates establishes an energy advantage. We also have no same-boundary measurement on a GPU with native four-bit floating-point support, so we make no GPU superiority claim.

VIII. Limitations and Threats to Validity

The long-horizon numerical evidence is layer-level and synthetic. The short Qwen diagnostic reconstructs query, key, value, decay, and delta-gate inputs at one pinned layer boundary, but contains only 12–18 valid tokens per prompt. It is not closed-loop execution and supports neither perplexity nor downstream-task claims. The kernel excludes projections, normalization, full-attention layers, and their key-value cache.

The official Qwen3-Next artifacts remain 80B-class models [4]. A bounded public Hugging Face API audit found no released Qwen3-Next recurrent activation capture and no smaller official trained checkpoint suitable for this host. Tiny-random and undocumented development checkpoints are not substitutes for real-model quality evidence.

The synthetic protocol covers a small registered seed set, two initial-state conditions, and one high-retention trace family. Metrics flatten heads, do not provide a trace-population confidence interval, and are not calibrated to perplexity or task accuracy. Replay and source hashes establish integrity, not external preregistration. Section VII gives an explanatory error recurrence, not a formal stability certificate.

The routed designs are all-layer recurrence-state subsystems synthesized out of context. They exclude the platform shell, weights, projections, convolution state, full-attention key/value caches, and host scheduling. The tested fixed-route points are not global frequency searches. Xilinx Runtime (XRT) execution, xclbin integration, physical board parity, telemetry, synchronized GPU measurements, and measured energy are absent. Vectorless power cannot substitute for any of them. A host and WSL inventory found no U55C XPFM, XRT management tools, or Xilinx PCI device.

Generated-RTL parity covers one registered synthetic trace. Full official XSIM recurrent cosimulation remains incomplete and is reported independently from the direct Verilator run. A completed official-XSim LOAD benchmark bounds the full trace at multiple days on this host, but is not parity evidence; neither path is used to fill missing board evidence. Results cover one U55C target, one P_K/P_V point, one block size, and batch one; multi-sequence occupancy, cross-device behavior, and service-level tail latency remain open.

IX. Conclusion

Native MXFP4 is feasible as an exact LUT-based arithmetic path for this persistent-state GDN recurrence core, but it is not a viable BF16 replacement under the joint criteria studied here. Uniform MXFP4 drifts; RS2/R3 meets the registered long synthetic stability bounds and reduces logical state storage, yet costs more LUTs and cycles and misses the target routed frequency. The result is therefore a qualified negative, not an accuracy or energy claim. Closed-loop model quality, shell-integrated board execution, and measured energy are required before revisiting that conclusion at model scale.

References

[1] S. Yang, J. Kautz, and A. Hatamizadeh, "Gated Delta Networks: Improving Mamba2 with Delta Rule," in *Proc. Int. Conf. Learning Representations*, 2025.

[2] N. Gupta, P. Wang, R. Kannan, and V. K. Prasanna, "A Persistent-State Dataflow Accelerator for Memory-Bound Linear Attention Decode on FPGA," arXiv preprint arXiv:2603.05931v1 [cs.AR], Mar. 2026, doi: 10.48550/arXiv.2603.05931. [Online]. Available: https://arxiv.org/abs/2603.05931

[3] Open Compute Project, "OCP Microscaling Formats (MX) Specification," ver. 1.0, Sep. 2023. [Online]. Available: https://www.opencompute.org/documents/ocp-microscaling-formats-mx-v1-0-spec-final-pdf

[4] Qwen Team, "Qwen3-Next-80B-A3B-Instruct," Hugging Face model repository, rev. 9c7f2fbe84465e40164a94cc16cd30b6999b0cc7, accessed Aug. 1, 2026. [Online]. Available: https://huggingface.co/Qwen/Qwen3-Next-80B-A3B-Instruct/tree/9c7f2fbe84465e40164a94cc16cd30b6999b0cc7

[5] Hugging Face, "Qwen3-Next implementation," *Transformers*, ver. 4.57.0, commit 8ac2b916b042b1f78b75c9eb941c0f5d2cdd8e10, 2025. [Online]. Available: https://github.com/huggingface/transformers/blob/8ac2b916b042b1f78b75c9eb941c0f5d2cdd8e10/src/transformers/models/qwen3_next/modeling_qwen3_next.py

[6] FLA Contributors, "Gated delta rule implementation," *flash-linear-attention*, ver. 0.2.1, commit a670dff4c2537fc1a82486584dd9569e18fba833, 2025. [Online]. Available: https://github.com/fla-org/flash-linear-attention/tree/a670dff4c2537fc1a82486584dd9569e18fba833/fla/ops/gated_delta_rule

[7] S. Yang, B. Wang, Y. Zhang, Y. Shen, and Y. Kim, "Parallelizing Linear Transformers with the Delta Rule over Sequence Length," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, doi: 10.52202/079017-3668.

[8] H. Wen, Z. Wang, J. Dong, W. Lou, L. Gong, C. Wang, and X. Zhou, "QLlama: An FPGA-Based Microscaling Quantization Accelerator for Energy-Efficient Llama2 Inference," *IEEE Embedded Systems Letters*, vol. 17, no. 5, pp. 337–340, 2025, doi: 10.1109/LES.2025.3600563.

[9] M. Mekhemer, A. Elsousy, B. Venkatesh, R. Rowley, V. Betz, N. Kapre, and A. Boutros, "Jack of All Scales: A Versatile FPGA Tensor Block for MXFP Precisions," arXiv preprint arXiv:2607.13898v1 [cs.AR], Jul. 2026, doi: 10.48550/arXiv.2607.13898.

[10] G. Karfakis *et al.*, "MXFormer: A Microscaling Floating-Point Charge-Trap Transistor Compute-in-Memory Transformer Accelerator," arXiv preprint arXiv:2602.12480v1 [cs.AR], Feb. 2026, doi: 10.48550/arXiv.2602.12480.

[11] W. Liu, H. Meng, Y. Luo, P. Zhang, and X. Ma, "MicroMix: Efficient Mixed-Precision Quantization with Microscaling Formats for Large Language Models," in *Proc. Int. Conf. Learning Representations*, 2026.

[12] J. Yu *et al.*, "MXAttention: Data-Free Optimal Scaling and Pre-Normalization Quantization for MXFP4 Attention," arXiv preprint arXiv:2607.24377v1, Jul. 2026, doi: 10.48550/arXiv.2607.24377.

[13] M. Zhang *et al.*, "Benchmarking Post-Training Quantization of Large Language Models under Microscaling Floating Point Formats," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics*, pp. 39907–39923, 2026, doi: 10.18653/v1/2026.acl-long.1854.

[14] L. Zhang *et al.*, "When Good Enough Is Optimal: Multiplication-Only Matrix Inversion Approximation for Quantized Gated DeltaNet," in *Proc. 43rd Int. Conf. Machine Learning*, PMLR, vol. 306, 2026.

[15] Z. Liu, J. Yuan, H. Jin, S. Zhong, Z. Xu, V. Braverman, B. Chen, and X. Hu, "KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache," in *Proc. 41st Int. Conf. Machine Learning*, PMLR, vol. 235, pp. 32332–32344, 2024.

[16] H. Kim, B. Ko, M. Kang, M. Kim, D. Lee, J. Lee, S. Yoon, and D. Jung, "SSDi8: Accurate and Efficient 8-bit Quantization for State Space Duality," in *Proc. Int. Conf. Learning Representations*, 2026.